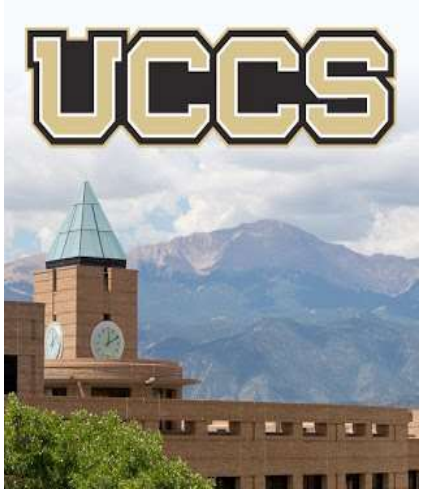


**Unit 01.03.01**
**CS 5220:**
**COMPUTER COMMUNICATIONS**

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# Berkeley Socket API - I

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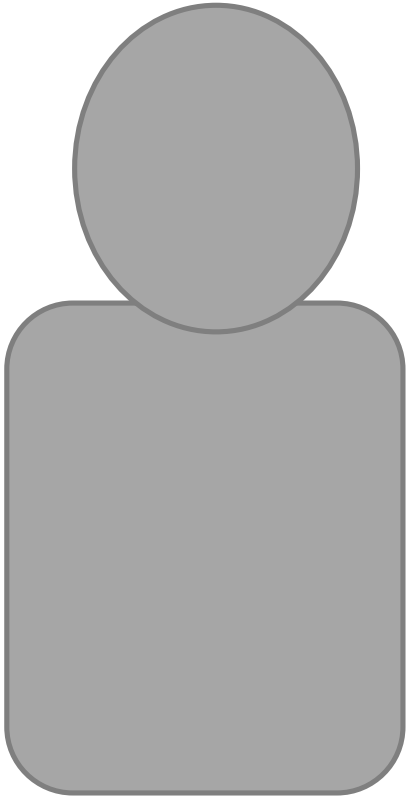
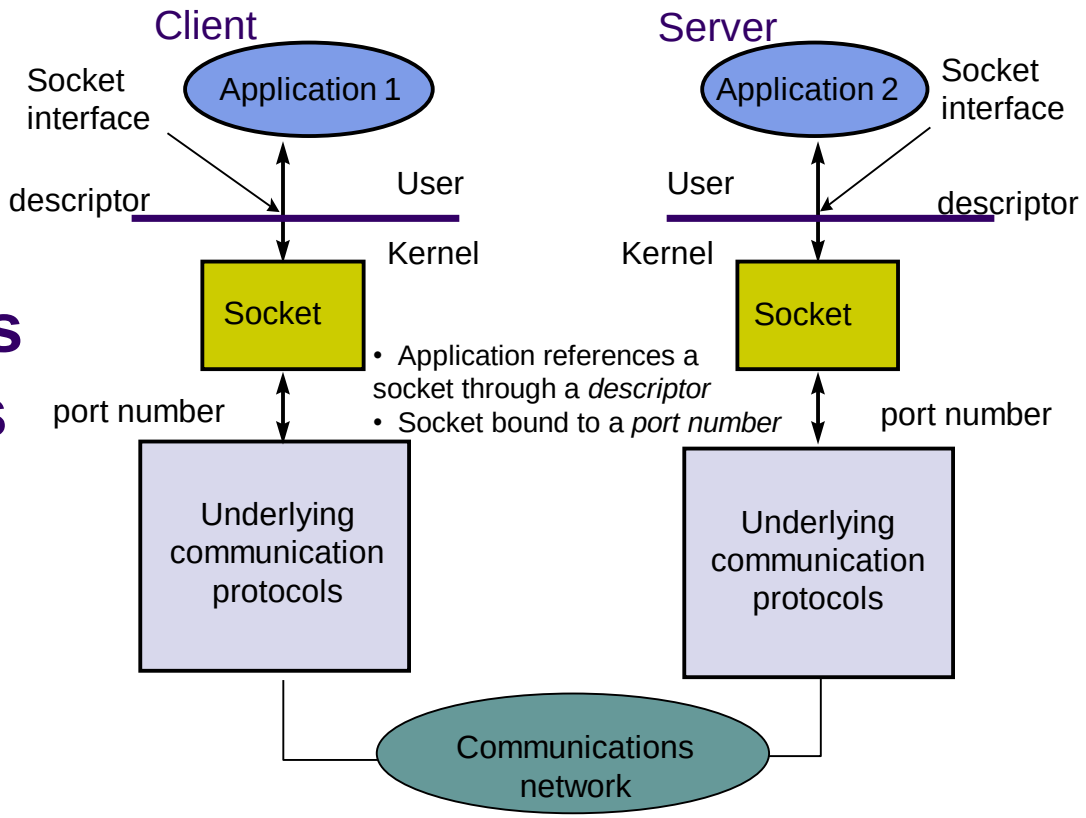
Professor, Department of Computer Science



# Berkeley Socket API

- Berkeley UNIX Sockets API
  - Abstraction for applications to send & receive data
  - Applications create sockets that “plug into” network
  - Applications write/read to/from sockets
  - Implemented in the kernel
  - Facilitates development of network applications
  - Hides details of underlying protocols & mechanisms
- Also in Windows, Linux, and other OS’s

# Communications through Sockets




- Application references a socket through a *descriptor*
- Socket bound to a *port number*

# Transport Protocols

- Host computers run two transport protocols on top of IP to enable process-to-process communications
- *User Datagram Protocol* (UDP) enables best-effort connectionless transfer of individual block of information
- *Transmission Control Protocol* (TCP) enables connection-oriented reliable transfer of a stream of bytes
- Two services though Sockets: connection-oriented and connection-less

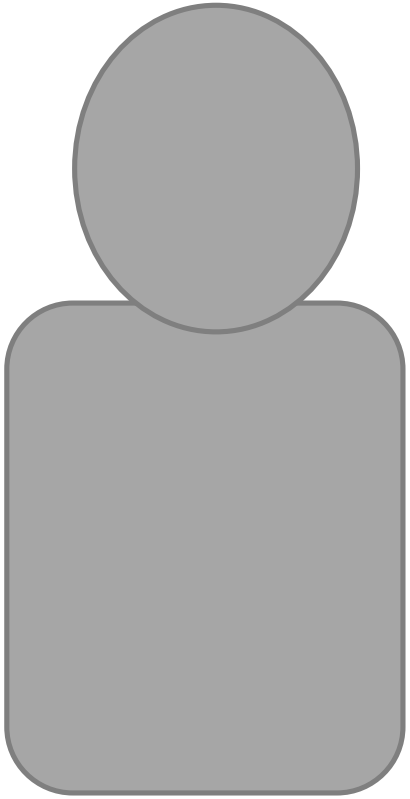
# Stream Mode of Service

Connection-oriented (TCP)

- First, setup connection between two peer application processes
- Then, reliable bidirectional in-sequence transfer of *byte stream* (boundaries not preserved in transfer)
- Multiple write/read between peer processes
- Finally, connection release

Connectionless (UDP)

- Immediate transfer of one block of information (boundaries preserved)
- No setup overhead & delay
- Destination address with each block
- Send/receive to/from multiple peer processes
- Best-effort service only
  - Possible out-of-order
  - Possible loss



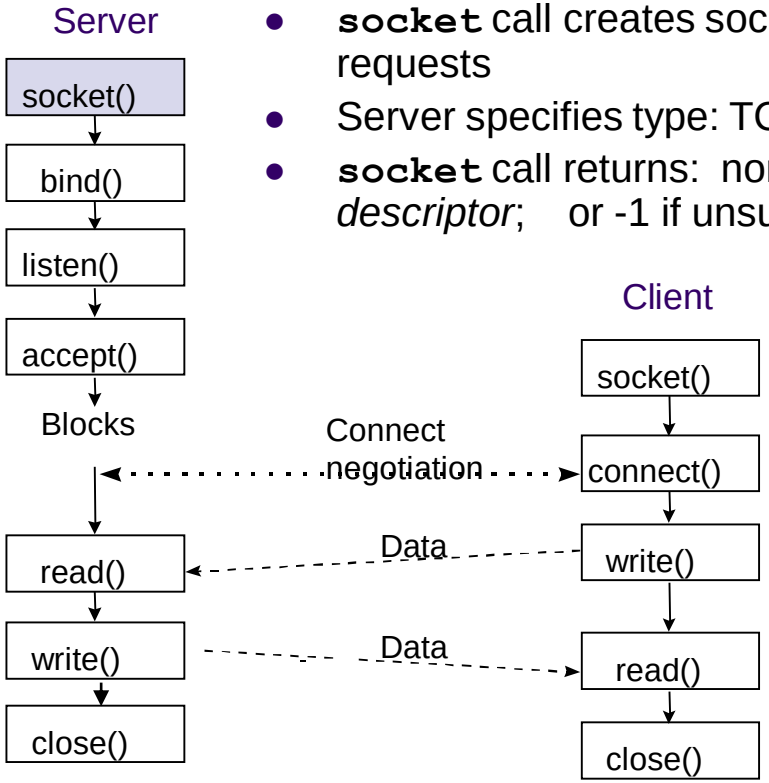
# Client & Server Differences

- Server
  - Specifies well-known port # when creating socket
  - May have multiple IP addresses (net interfaces)
  - Waits passively for client requests
- Client
  - Assigned ephemeral port #
  - Initiates communications with server
  - Needs to know server's IP address & port #
    - DNS for URL & server well-known port #
  - Server learns client's address & port #

# Socket Calls for Connection-Oriented Mode

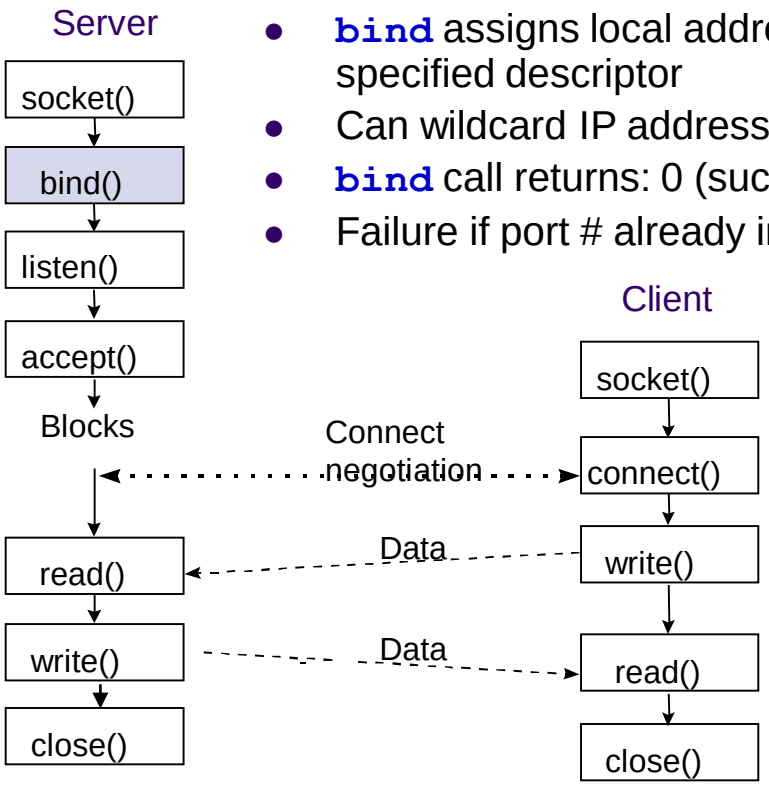
Server does *Passive* Open

- **`socket`** call creates socket to *listen* for connection requests
- Server specifies type: TCP (stream)
- **`socket`** call returns: non-negative integer *descriptor*; or -1 if unsuccessful



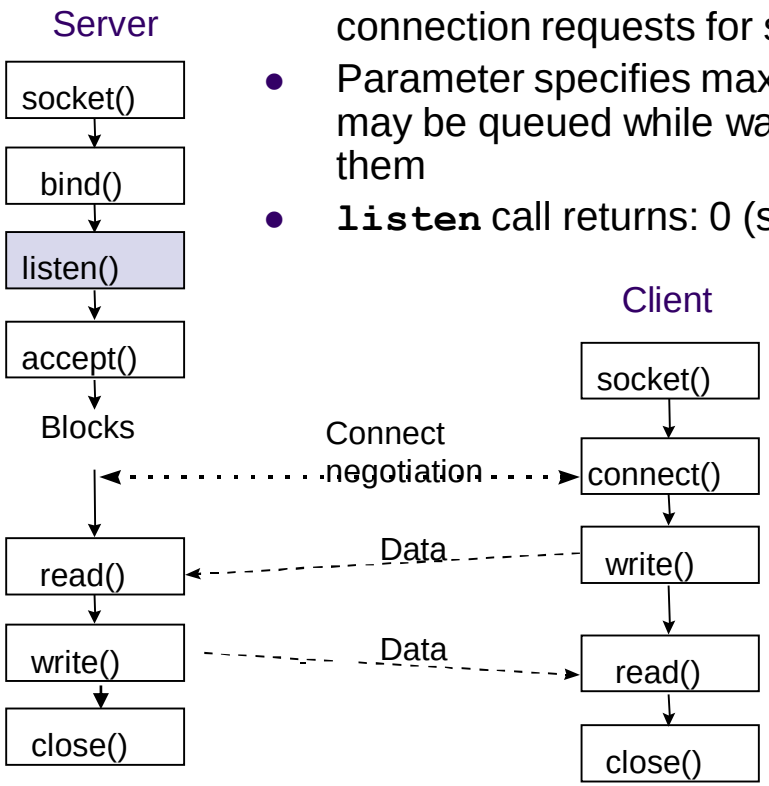
# Socket Calls for Connection-Oriented Mode

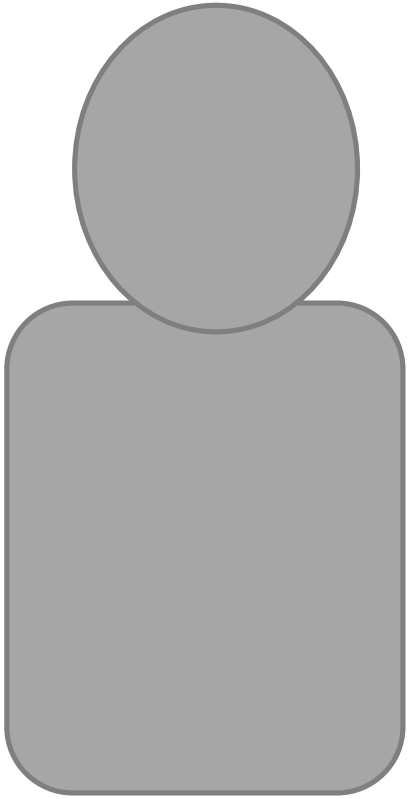
- **`bind`** assigns local address & port # to socket with specified descriptor
- Can wildcard IP address for multiple net interfaces
- **`bind`** call returns: 0 (success); or -1 (failure)
- Failure if port # already in use or if reuse option not set

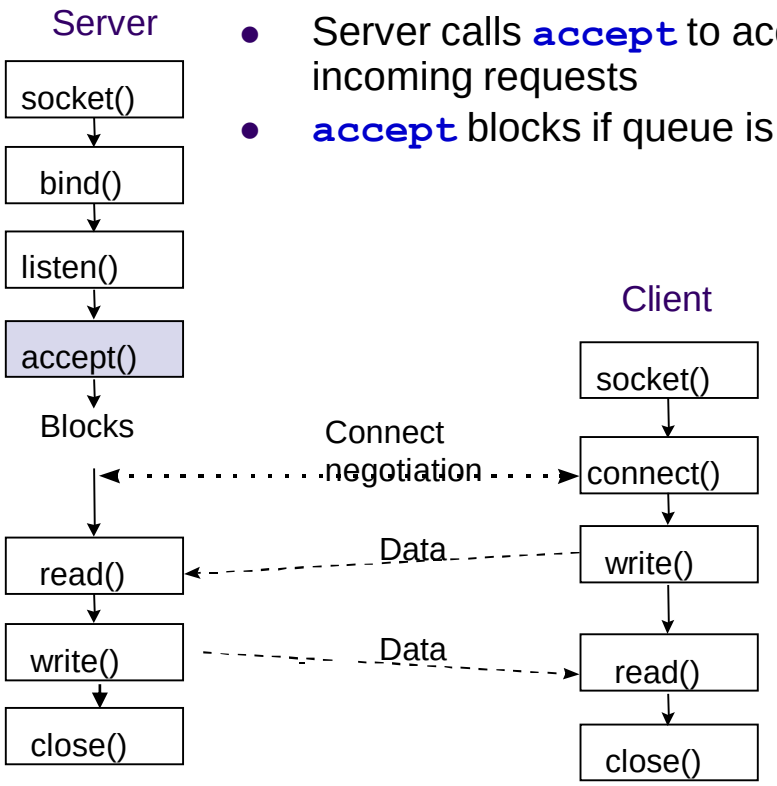


# Socket Calls for Connection-Oriented Mode

- **`listen`** indicates to TCP readiness to receive connection requests for socket with given descriptor
- Parameter specifies max number of requests that may be queued while waiting for server to accept them
- **`listen`** call returns: 0 (success); or -1 (failure)





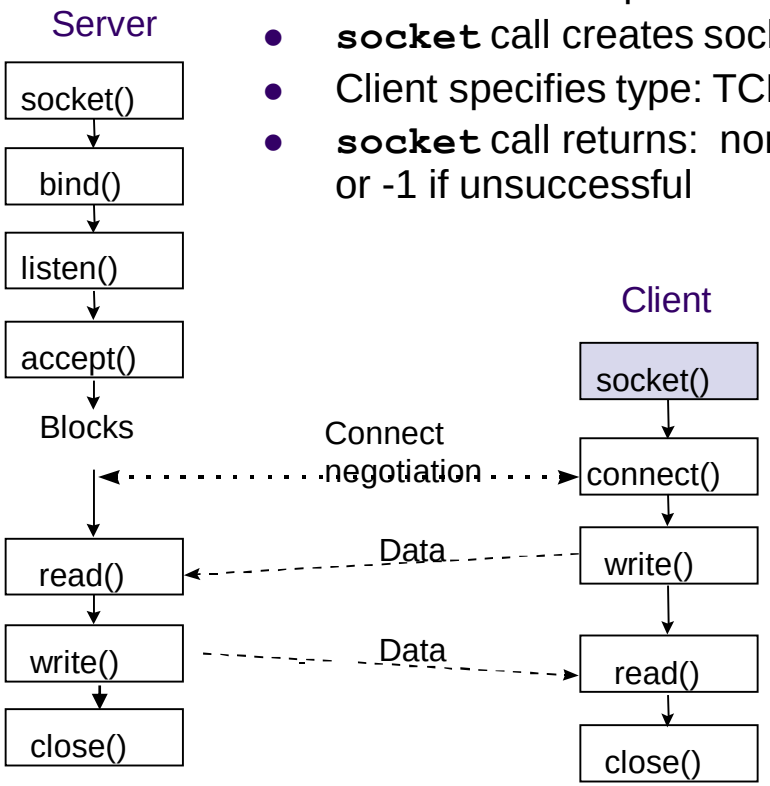



- Server calls `accept` to accept incoming requests
- `accept` blocks if queue is empty

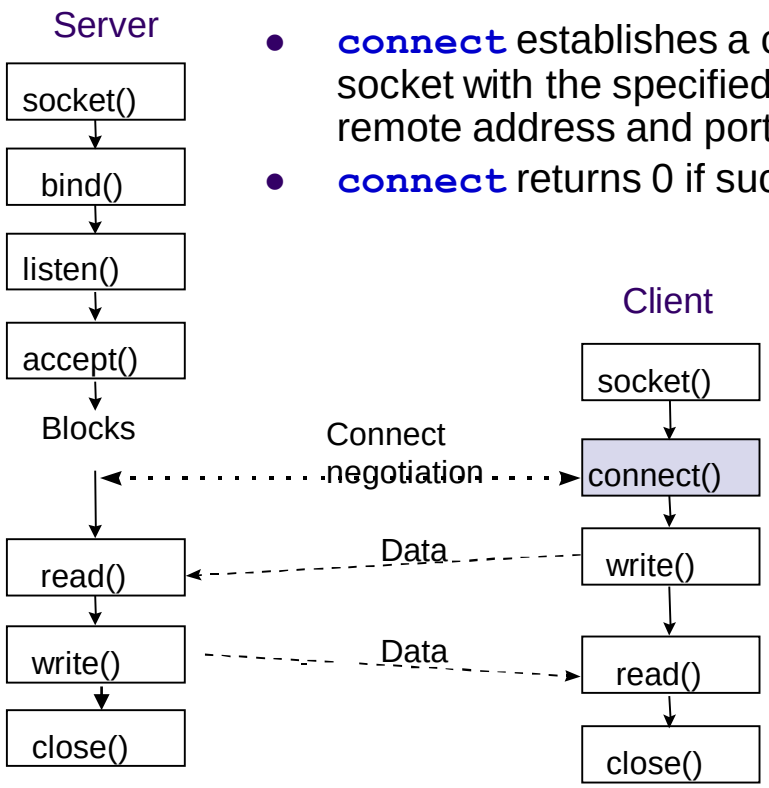
# Socket Calls for Connection-Oriented Mode

Client does Active Open

- **`socket`** call creates socket to connect to server
- Client specifies type: TCP (stream)
- **`socket`** call returns: non-negative integer *descriptor*; or -1 if unsuccessful



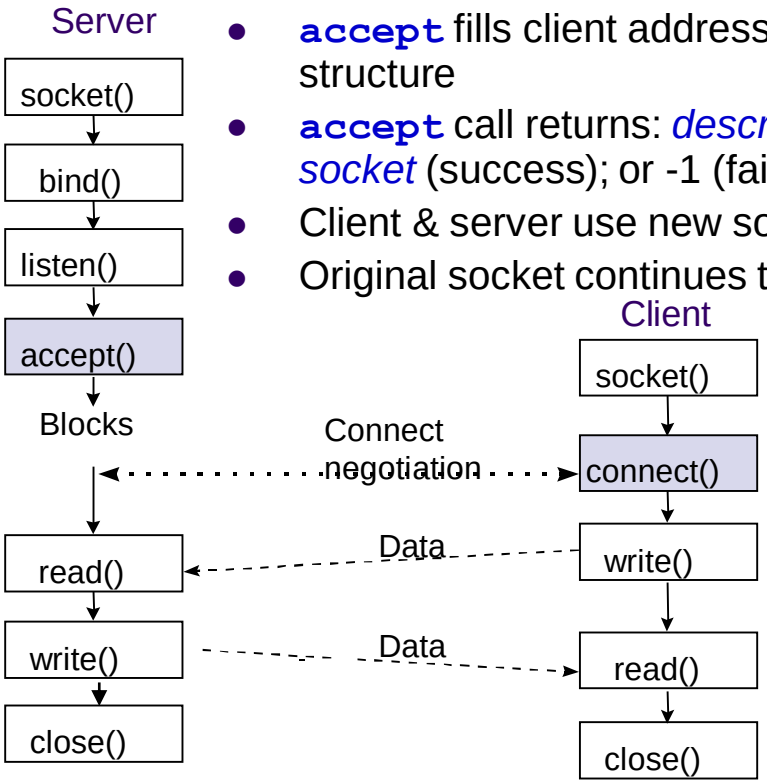
- **`connect`** establishes a connection on the local socket with the specified descriptor to the specified remote address and port #
- **`connect`** returns 0 if successful; -1 if unsuccessful

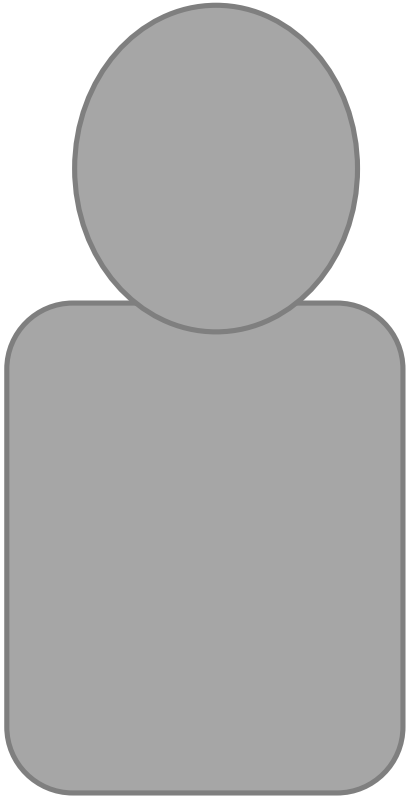



Note: **`connect`** initiates TCP three-way handshake

# Socket Calls for Connection-Oriented Mode

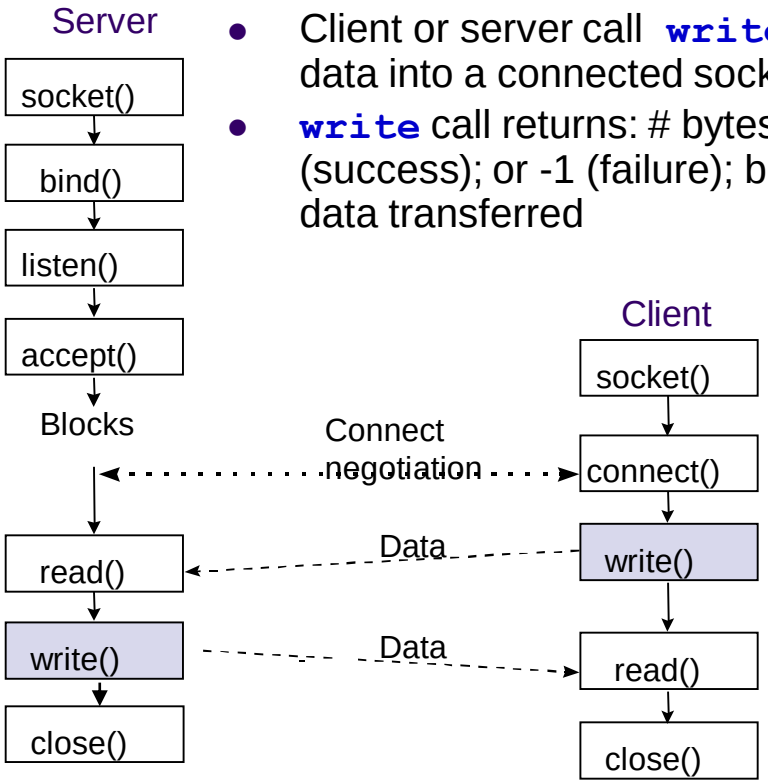
- **`accept`** wakes with incoming connection request
- **`accept`** fills client address & port # into address structure
- **`accept`** call returns: *descriptor of **new** connection socket* (success); or -1 (failure)
- Client & server use new socket for data transfer
- Original socket continues to listen for new requests





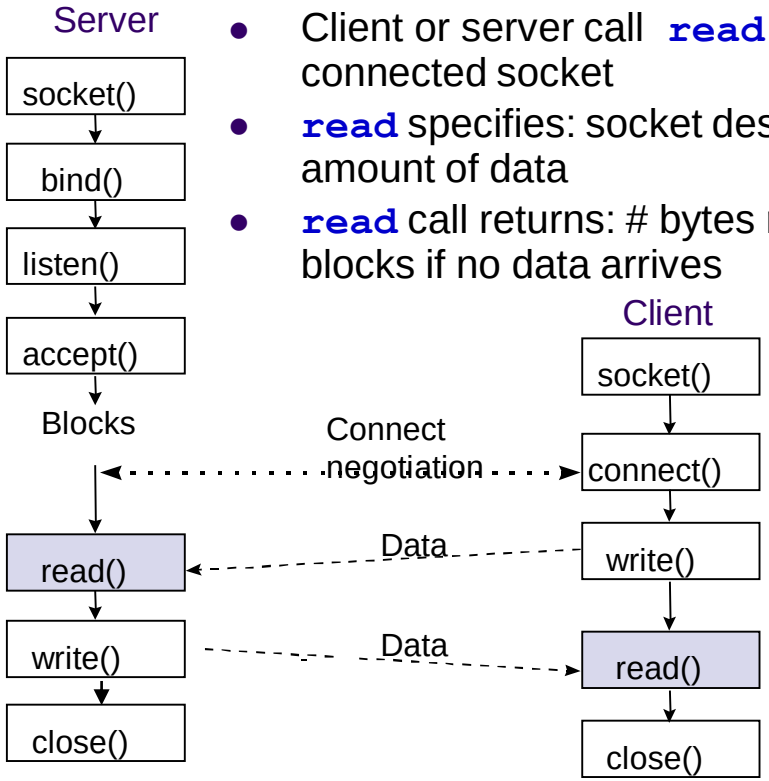
## Data Transfer

- Client or server call `write` to transmit data into a connected socket
- `write` call returns: # bytes transferred (success); or -1 (failure); blocks until all data transferred



## Data Transfer

- Client or server call **`read`** to receive data from a connected socket
- **`read`** specifies: socket descriptor; pointer to a buffer; amount of data
- **`read`** call returns: # bytes read (success); or -1 (failure); blocks if no data arrives

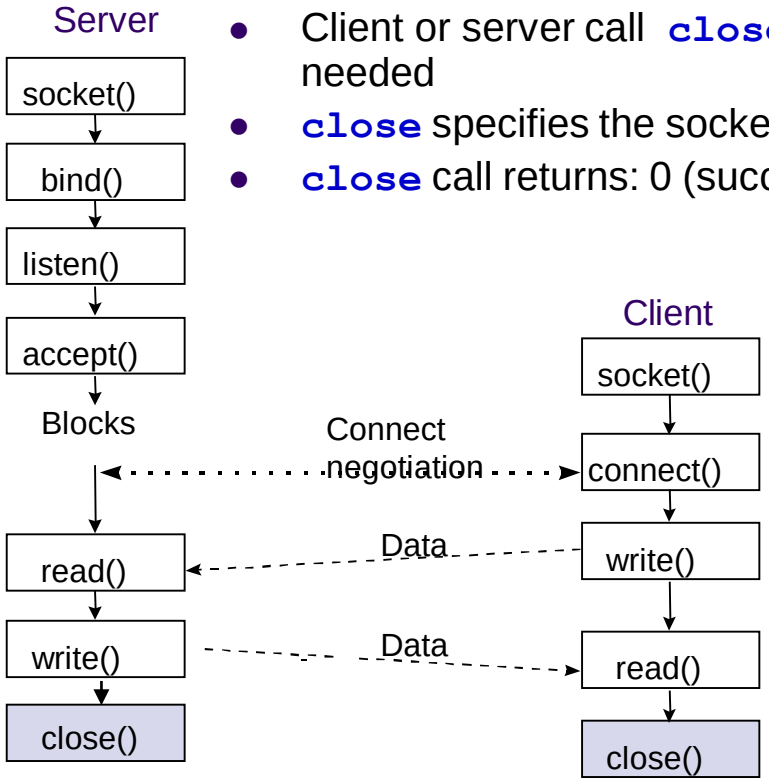



Note: **`write`** and **`read`** can be called multiple times to transfer byte streams in both directions

# Socket Calls for Connection-Oriented Mode

## Connection Termination

- Client or server call `close` when socket is no longer needed
- `close` specifies the socket descriptor
- `close` call returns: 0 (success); or -1 (failure)




Note: `close` initiates TCP graceful close sequence

# Summary of the Lesson

- Socket API hides details of underlying protocols & mechanisms